

Cyber Forensics Final Solution

Module 1: Introduction to Digital Forensics

1. **[Mid 1] Define digital forensics? [1 Mark]**

Digital forensics is the use of tools and techniques to recover, preserve, and examine digital evidence from digital devices.

2. **[Mid 1] Define cybercrime. OR Define Cyber Crime. [1 Mark]**

Cybercrime is any illegal activity carried out using computers, networks, or the internet.

3. **[Mid 1] Define Computer Vandalism. [1 Mark]**

Computer vandalism is the intentional destruction or damage of computer systems, hardware, or data belonging to others.

4. **[Mid 1] What is violent cybercrime? [1 Mark]**

Violent cybercrime refers to cyber activities that can cause physical risk or harm to individuals, such as cyber terrorism or threats.

5. **[Mid 1] Define Malware. [1 Mark]**

Malware is malicious software designed to secretly access or damage a computer system, usually to steal data.

6. **[Mid 1] Give the full form of CSIRT. [1 Mark]**

Computer Security Incident Response Team.

7. **[Mid 1] What is Software Piracy? [1 Mark]**

Software piracy is the unauthorized copying, distribution, or use of software.

8. **[Mid 1] Give any two digital forensic goals. [1 Mark]**

1. Recover and preserve digital evidence.
2. Analyze evidence to reconstruct events or identify suspects.

9. [Mid 1] Define CACHE and RAM. [1 Mark]

- **Cache:** A small, highly volatile memory that stores recently used data for quick access.
- **RAM:** Main memory that stores currently running programs and data temporarily.

10. [Mid 2] Digital forensics is mainly used to: [1 Mark]

- (a) Design software
- (b) Investigate cybercrimes**
- (c) Develop websites
- (d) Install operating systems

11. [Mid 1] List and explain any five cybercrimes. [5 Marks]

1. **Hacking:** Unauthorized access to a computer system or network to steal, modify, or destroy data.
2. **Malware Attacks:** Use of malicious software (viruses, worms, trojans) to damage systems, steal data, or gain unauthorized access.
3. **Software Piracy:** Illegal copying, distribution, or use of copyrighted software without permission.
4. **Cyber Stalking:** Tracking or harassing a person online through emails, messages, or social media, often with threats.
5. **Computer Vandalism:** Deliberate destruction or damage of computer systems, data, or hardware belonging to others.

12. [Mid 1] Explain Integrated Digital Investigation Process Model (IDIP). [5 Marks]

The **Integrated Digital Investigation Process (IDIP) model**, proposed by Carrier and Spafford in 2003, integrates physical and digital crime scene investigations into a single framework.

Phases:

1. **Readiness:** Preparing infrastructure, tools, and trained personnel to handle future investigations.
2. **Deployment:** Detecting and confirming incidents and obtaining authorization to investigate.
3. **Physical Crime Scene Investigation:** Collecting and preserving physical evidence related to the crime.
4. **Digital Crime Scene Investigation:** Identifying, collecting, examining, and analyzing digital evidence.
5. **Review:** Evaluating the entire investigation to identify lessons learned and improve future processes.

13. [Mid 1] What are the various types of the Digital Forensics? [3 Marks]

- **Computer Forensics:** Analysis of data from computers and storage devices like hard disks and USB drives.
- **Mobile Device Forensics:** Extraction of evidence from smartphones such as calls, messages, and app data.
- **Network Forensics:** Monitoring and analysis of network traffic to detect intrusions and unauthorized access.
- **Database Forensics:** Examination of databases to identify data manipulation or unauthorized access.
- **Memory Forensics:** Analysis of volatile data from RAM of a running system.
- **Multimedia Forensics:** Analysis of images, audio, and video to check authenticity and detect tampering.

14. [Mid 1] What is CSIRT? Explain its functions. [4 Marks]

CSIRT (Computer Security Incident Response Team) is a specialized group of professionals responsible for preventing, detecting, analyzing, responding to, and recovering from computer security incidents within an organization or at a national level.

Functions of CSIRT:

1. **Incident detection and reporting:** Monitors systems and networks to identify security incidents and provides a mechanism for users to report incidents.
2. **Incident analysis and assessment:** Analyzes the nature, scope, severity, and impact of incidents such as malware infections, intrusions, or data breaches.
3. **Incident response and containment:** Takes immediate actions to contain incidents, isolate affected systems, and limit further damage.
4. **Incident recovery:** Assists in restoring systems, services, and data to normal operation after an incident.
5. **Evidence collection and preservation:** Collects and preserves digital evidence in a forensically sound manner to support investigations and legal actions.
6. **Vulnerability management:** Identifies security weaknesses and recommends patches, configuration changes, or mitigation strategies.
7. **Post-incident analysis and improvement:** Reviews incidents to identify lessons learned and improves policies, procedures, and defenses to prevent future incidents.

15. List and explain different types of the categories of the cyber crime. **OR** Explain different types of cyber-crimes.

1. **Cyber Crimes Against Persons:** Crimes committed against individuals using computers or the Internet and mainly affect personal safety, privacy, and reputation.

Examples:

1. Cyber stalking
2. Cyber harassment
3. Email spoofing
4. Identity theft

2. **Cyber Crimes Against Property:** Crimes which target digital or physical property related to computers and networks.

Examples:

1. Computer vandalism
2. Malware attacks
3. Software piracy
4. Cyber fraud and theft

3. **Cyber Crimes Against Government:** Crimes directed against government organizations, systems, or national security.

Examples:

1. Cyber terrorism
2. Hacking government systems
3. Denial-of-Service (DoS) attacks

16. What are digital forensics?

Digital forensics refers to the use of specialized tools and techniques to identify, collect, preserve, examine, and analyze digital evidence legally.

It involves recovering evidence from digital devices and electronic media such as computers, laptops, mobile phones, servers, storage devices, and networks, including data that may be deleted, hidden, or encrypted.

The main objective of digital forensics is to reconstruct events, support criminal investigations, identify suspects, and present reliable digital evidence in a court of law.

17. Explain Violent and Nonviolent cyber-crimes.

Violent Cyber Crimes: Cyber crimes which directly or indirectly pose a physical risk to individuals, society, or national security. They may cause fear, harm, or loss of life. These crimes are considered serious because they may escalate into real-world violence.

Examples:

- Cyber terrorism
- Cyber stalking
- Online threatening
- Child pornography

Non-Violent Cyber Crimes: Cyber crimes which do not cause direct physical harm but result in financial loss, data loss, privacy invasion, or damage to reputation. Although non-violent, these crimes can cause serious economic and personal damage.

Examples:

- Cyber theft
- Cyber fraud
- Cyber trespass
- Destructive cyber crimes

18. What is a role of computer forensics in the crime?

1. **Computer as the Object of Crime:** When a computer itself is targeted by a criminal act, such as theft, damage, or destruction of a computer system. Computer forensics helps examine the system to determine how and when the crime occurred.
2. **Computer as the Subject of Crime:** When a computer is the environment in which the crime is committed, for example when a system is infected by a virus, worm, or malware. Computer forensics is used to trace the source and impact of the attack.
3. **Computer as the Tool of Crime:** When a computer is used to plan or commit a crime, such as hacking, online fraud, identity theft, or document forgery. Forensic analysis helps recover logs, files, emails, and other digital traces to identify the offender.
4. **Computer as a Symbol of Crime:** When the image or claim of computer use is employed to deceive or intimidate victims, such as false claims of using powerful systems or software to commit fraud. Computer forensics helps verify the truth behind such claims.

19. Why digital forensics required?

1. **Recovery of hidden or deleted data:** Deleted files, formatted disks, and uninstalled applications often leave traces. Digital forensics helps recover such data.
2. **Preservation of digital evidence:** It ensures evidence is collected and preserved without alteration, maintaining its integrity and legal validity.
3. **Reconstruction of events:** Digital forensics helps recreate timelines of activities to understand how, when, and where a crime occurred.
4. **Identification of suspects:** Analysis of logs, emails, network traffic, and user activity helps trace actions back to individuals.
5. **Support for legal proceedings:** Proper forensic methods make digital evidence admissible in court.
6. **Handling volatile data:** It enables collection of volatile data such as RAM, cache, and running processes that would otherwise be lost.
7. **Support for cybersecurity and incident response:** Digital forensics helps analyze breaches, malware attacks, and insider threats to prevent future incidents.

20. Digital Forensics Research Workshop Investigative Model (DFRWS).

The **Digital Forensics Research Workshop Investigative (DFRWS) model** was proposed in 2001 and is one of the **earliest and most widely accepted digital forensic investigation frameworks**. It provides a structured and systematic approach to handling digital evidence. This model emphasizes evidence integrity and legal acceptability.

Phases:

1. **Identification:** Detecting an incident and identifying potential sources of digital evidence through monitoring, audits, or alerts.
2. **Preservation:** Ensuring that identified evidence is protected from alteration or damage. This includes maintaining chain of custody.
3. **Collection:** Gathering digital evidence using approved forensic methods and tools.
4. **Examination:** Processing collected data to extract relevant information, including recovery of deleted or hidden files.
5. **Analysis:** Interpreting examined data to reconstruct events, determine timelines, and identify suspects.
6. **Presentation:** Documenting findings and presenting them in reports or court testimony.

21. Explain Enhanced Digital Investigation Process Model (EDIP).

The **Enhanced Digital Investigation Process (EDIP)** model is an enhancement of the IDIP model. It introduces additional phases to better handle complex cyber crimes, especially those involving remote attackers.

Phases:

1. **Readiness:** Same as IDIP, ensuring tools and personnel are prepared.
2. **Deployment:** Detection, confirmation, and authorization of investigation.
3. **Traceback:** Tracing the attack back to the originating system, device, or network.
4. **Dynamite (Investigation Phase):** Conducting detailed physical and digital investigations at the primary crime scene to identify suspects.
5. **Review:** Assessing investigation effectiveness and improving procedures.

22. Explain Computer Forensics Field Triage Process Model (CFFTPM).

The **Computer Forensics Field Triage Process Model (CFFTPM)** focuses on rapid, on-site forensic analysis rather than full laboratory investigations. This model is **useful when time is limited and quick decisions are required** at the crime scene.

Phases:

1. **Planning:** Preparing for the investigation and defining objectives.
2. **Triage:** Identifying and prioritizing the most critical and volatile evidence.
3. **User Usage Profile:** Analyzing user activity to associate actions with suspects.
4. **Chronology Timeline:** Reconstructing events using timestamps and system logs.
5. **Internet Phase:** Examining internet-related artifacts such as browser history, emails, and chat logs.
6. **Case-Specific Evidence:** Focusing on evidence directly relevant to the specific crime.

23. List different digital forensic models.

- Digital Forensics Research Workshop (DFRWS) Investigative Model
- Integrated Digital Investigation Process Model (IDIP)
- Enhanced Digital Investigation Process Model (EDIP)
- Computer Forensics Field Triage Process Model (CFFTPM)
- Common Phases of Computer Forensic Investigation Models (CPCCFIM)

24. Explain Discuss Common phases of computer forensic investigation model (CPCCFIM).

The **Computer Forensic Investigation Model (CPCCFIM)** represents common stages found across many forensic investigation frameworks. This model provides a generalized and flexible approach applicable to most digital forensic investigations.

Phases:

1. **Pre-Process:** Obtaining authorization, preparing tools, and planning the investigation.
2. **Acquisition and Preservation:** Collecting digital evidence and ensuring it remains unaltered.
3. **Analysis:** Examining and interpreting data to uncover relevant evidence and reconstruct events.
4. **Presentation:** Reporting findings in a clear and legally acceptable format.
5. **Post-Process:** Closing the investigation, returning evidence, and reviewing procedures.

25. What are the various objectives of a Digital Forensic process? **OR** What is the overall Digital Forensic process?

1. **Identification:** Detect the incident and identify potential sources of digital evidence such as computers, mobile devices, networks, and storage media.
2. **Preservation:** Protect evidence from alteration or damage by maintaining integrity and proper chain of custody.
3. **Collection (Acquisition):** Collect digital evidence using forensically sound methods, like bit-by-bit imaging, without modifying the original data.
4. **Examination:** Extract and recover relevant data such as deleted, hidden, or encrypted files.
5. **Analysis and Reconstruction:** Analyze data to reconstruct events, determine what happened, how it happened, and identify individuals involved.
6. **Documentation and Presentation:** Record procedures and present findings clearly so they are understandable and legally admissible.
7. **Review and Closure:** Review the investigation, improve processes, and securely close the case.
8. **Preservation of Findings:** Store reports, evidence, and documentation securely for future reference, legal proceedings, or re-investigation.

26. What are the goals of Incidence response? **OR** What happens after detecting an incident?

1. **Detection and identification:** Recognize and confirm the incident, then classify it based on type and severity.
2. **Containment:** Isolate affected systems or networks to prevent further damage.
3. **Evidence preservation:** Collect and preserve digital evidence while maintaining proper chain of custody.
4. **Eradication:** Identify and remove the root cause such as malware or unauthorized access.
5. **Recovery:** Restore systems, data, and services to normal operation safely.
6. **Notification and escalation:** Inform relevant stakeholders and escalate if necessary.
7. **Post-incident review and prevention:** Analyze the incident, minimize losses, and improve security measures to prevent future incidents.

Module 2: Digital Evidence, Forensic Duplication and Digital Evidence Acquisition

1. [Mid 1] Define Non-volatile Data. [1 Mark]

Non-volatile data is data that remains stored even after the system is powered off, such as data on hard drives or storage media.

2. [Mid 2] A forensic image refers to: [1 Mark]

- (a) Screenshot of a file
- (b) Exact copy of digital evidence**
- (c) Compressed file
- (d) Photo of computer

3. [Mid 2] Chain of custody refers to: [1 Mark]

- (a) List of suspects
- (b) Record of evidence handling**
- (c) List of investigators
- (d) Crime location report

4. [Mid 2] Which device can store digital evidence? [1 Mark]

- (a) Hard disk
- (b) USB drive
- (c) Mobile phone
- (d) All of the above**

5. [Mid 2] Hash values are used to: [1 Mark]

- (a) Encrypt files
- (b) Verify data integrity**
- (c) Compress data
- (d) Delete data

6. **[Mid 1]** What is chain of custody? **OR** Explain “Chain of Custody” process in detail. **OR** Explain the process of the chain of custody in detail. **[5 Marks]**

Chain of custody is the chronological documentation and record of the collection, handling, transfer, analysis, and storage of evidence to ensure its integrity and admissibility in court. It records who handled the evidence, when, where, and why, and prevents tampering, making the evidence legally acceptable.

Chain of Custody Process:

1. **Data Collection:** Evidence is identified, labeled, documented, and collected from the source with details like date, time, and method while preserving integrity.
 2. **Examination:** Evidence is examined using forensic tools without altering original data, and all actions are documented with logs or screenshots.
 3. **Analysis:** Relevant information is extracted by reconstructing events, recovering deleted data, and interpreting findings using valid methods.
 4. **Reporting:** The entire process, tools used, and conclusions are documented for legal purposes.
 5. **Storage and Preservation:** Evidence is securely stored in sealed and labeled containers with restricted access to prevent tampering.
 6. **Presentation in Court:** Evidence and reports are presented, and chain of custody proves authenticity with possible expert testimony.
 7. **Final Disposition:** After case completion, evidence is returned or disposed of as per legal requirements.
7. **[Mid 1]** List and explain types of the formats of a digital image. **OR** Explain different forensic image and format types. **[3 Marks]**
- **RAW (dd format):** A bit-by-bit copy of the storage device with no compression; simple, fast, and widely supported.
 - **E01 (EnCase format):** A compressed forensic image format that stores metadata, supports integrity checks (hashes), and allows splitting into segments.
 - **EX01:** An advanced version of E01 with improved compression and support for larger storage devices.
 - **AFF (Advanced Forensic Format):** An open-source format that supports compression, metadata storage, and efficient evidence handling.

8. **[Mid 1]** Explain forensic cloning in detail. **OR** What is forensic cloning? **OR** What is disk imaging? **[3 Marks]**

Forensic cloning (disk imaging) is the process of creating an exact bit-by-bit copy of a digital storage device such as a hard disk, SSD, or USB drive. It captures all data including active files, deleted files, hidden data, file system structures, metadata, and unallocated space, ensuring no potential evidence is missed. The original device is preserved as evidence, and all analysis is performed on the cloned image to maintain integrity and legal admissibility. Hash values such as MD5 or SHA are generated to verify that the cloned copy is identical to the original.

9. **[Mid 1]** What is a Digital Evidence? What are its various types? **OR** What is a digital evidence? Give examples. **[4 Marks]**

Digital evidence is any legally acceptable evidence that is stored, transmitted, or processed in digital form and can be used to prove or disprove facts in an investigation or court of law. It includes data found on electronic devices or generated during digital activities and must be collected and preserved in a forensically sound manner.

Examples:

- System and event logs.
- Emails and email attachments.
- Chat messages and social media conversations.
- Call logs, SMS, and app data from mobile phones.
- Deleted or hidden files recovered from storage media.
- Internet artifacts like browser history, cookies, and cache.
- USB drives, hard disks, memory cards, and other digital storage media.
- Computer files such as documents, spreadsheets, images, audio, and videos.

10. What are the various obstacles/challenges in evidence handling? **OR** What are the various challenges in acquiring digital evidences?

- **Maintaining chain of custody:** Any break in documentation or improper handling can make evidence legally inadmissible.
- **Risk of evidence tampering or alteration:** Digital evidence is fragile and can be easily modified, intentionally or accidentally, during collection or analysis.
- **Handling volatile data:** Volatile data such as RAM, cache, and running processes can be lost if the system is powered off or not captured in time.
- **Large volume of data:** Modern storage devices contain massive amounts of data, making identification and analysis time-consuming.
- **Encryption and passwords:** Strong encryption and unknown passwords make accessing evidence difficult.

11. Explain the role of admissibility of evidence.

Admissibility ensures that only legally valid and reliable evidence is accepted in court. It requires proper collection, preservation, and handling of evidence to maintain integrity and prevent tampering. It also verifies authenticity and credibility, ensuring fair trials and accurate judgments, while excluding any illegally obtained evidence.

12. List and explain other types of the evidences.

1. **Demonstrative (Illustrative) Evidence:** Evidence which visually demonstrates or explains facts to make them easier to understand.
 - **Examples:** Photographs, videos, audio recordings, maps, charts, graphs, models, simulations, x-rays.
2. **Documentary Evidence:** Consists of information presented in written or recorded form.
 - **Examples:** Contracts, invoices, wills, letters, emails, printed documents, digital documents, photographs, and recordings.
3. **Exculpatory (Explainable) Evidence:** Evidence which helps to reduce or eliminate guilt of the accused.
 - **Examples:** Alibi records, logs proving absence from a crime scene, messages or documents that contradict allegations.
4. **Substantial (Physical) Evidence:** Tangible evidence that can be directly examined.
 - **Examples:** Weapons, fingerprints, blood samples, DNA, hard drives, mobile phones, damaged devices.
5. **Scientific Evidence:** Evidence derived using scientific methods and expert analysis and must meet accepted scientific standards.
 - **Examples:** DNA analysis, fingerprint analysis, forensic reports, digital forensic analysis results.
6. **Testimonial Evidence:** Verbal or written evidence given by a witness under oath.
 - **Examples:** Statements made in court, depositions, affidavits, expert testimony.

Module 3: Forensic Investigation

1. **[Mid 2]** Which tool is commonly used for digital analysis? **[1 Mark]**
 - (a) Photoshop
 - (b) Autopsy**
 - (c) MS Word
 - (d) PowerPoint

2. **[Mid 2]** What are the 5 rules of the evidence? Explain in detail. **OR** Discuss the five rules of evidence. **[5 Marks]**
 1. **Admissible:** Evidence must be legally obtained and allowed in court according to the law. If the evidence is collected without proper authorization or violates legal procedures, it may be rejected by the court.
 2. **Authentic:** Evidence must be proven to be genuine and related to the incident under investigation. The investigator must demonstrate that the evidence has not been altered and that it originated from the specific device or system being examined.
 3. **Complete:** Evidence must present the entire picture of the incident. Investigators should collect all relevant data so that the court can clearly understand what actually happened.
 4. **Reliable:** Evidence must be collected and analyzed using reliable forensic tools and methods. The procedures used should produce consistent and accurate results so that the evidence can be trusted.
 5. **Believable:** Evidence must be clear and understandable to the court or jury. The investigator should present the findings in a way that non-technical people can easily understand.

3. What are the various hard drive analysis techniques?
 1. **Data Carving:** Raw data is searched for file signatures to recover files, usually from unallocated space, even when file metadata is missing.
 2. **EnCase:** In EnCase, investigators can use the “Deleted Files” filter or File Finder to locate deleted files or specific file types.
 3. **FTK Imager:** FTK Imager can recover deleted files by adding the physical drive as evidence, locating the file in the evidence tree, and exporting it to another location.
 4. **Fatback (Linux):** Fatback is a Linux forensic tool used to recover deleted files from disk images or storage devices.
 5. **Foremost (Linux):** Foremost is a recovery tool that restores files by searching for file headers and footers in forensic images.

4. What are the various router investigation techniques?

1. **Establishing a Router Connection:** The investigator first establishes a direct connection to the router to collect volatile information without alerting an attacker on the network.
2. **Recording System Time:** After connecting, the router's system time is recorded to help correlate logs and evidence later.
Example command: show clock
3. **Determining Who is Logged On:** The investigator checks currently logged-in users to identify any unauthorized access.
Example command: show users
4. **Reviewing the Routing Table:** The routing table is examined to understand how the router forwards packets and to detect any suspicious routing changes.
Example command: show ip route
5. **Checking Interface Configurations:** The router's network interface settings are reviewed to identify abnormal configurations.
Example command: show ip interface

5. What are the various malware analysis techniques?

1. **PeStudio:** Used for initial malware triage to identify suspicious artifacts, generate file hashes, check detections with VirusTotal, extract strings, domains, and IP addresses, and analyze entropy to detect packed or obfuscated malware.
2. **Process Hacker:** Displays running processes on a system and helps detect processes created by malware while allowing analysts to inspect process memory for strings such as IP addresses, domains, and user agents.
3. **Process Monitor (ProcMon):** Records real-time system activity such as file operations, process creation, and registry changes, helping analysts identify executed programs and their parent-child relationships.
4. **Autoruns:** Lists programs that run automatically during system startup and helps detect malware persistence mechanisms and new startup entries.
5. **Ghidra:** A reverse engineering tool that works as a disassembler and decompiler to convert compiled malware code into human-readable form.

6. What are the Forensic Artifacts?

Forensic artifacts are pieces of digital information or traces left behind by user activities and system processes on a computer system. These artifacts are stored in locations such as the file system, Windows registry, log files, browser history, and application data. During a forensic investigation, analysts examine these artifacts to reconstruct events, track user actions, detect suspicious activity, and collect digital evidence related to an incident.

7. State the difference between investigation and forensics.

Investigation	Forensics
Investigation is the process of searching for facts and information related to a crime or incident.	Forensics is the scientific analysis of evidence collected during an investigation.
It focuses on identifying suspects, motives, and events .	It focuses on examining and analyzing evidence using scientific methods.
It may involve interviews, surveillance, and gathering information .	It involves technical examination of digital or physical evidence .
The goal is to understand what happened and who is responsible .	The goal is to produce reliable evidence that can be presented in court .
Investigation is a broader process that includes many activities.	Forensics is a specialized part of the investigation process .

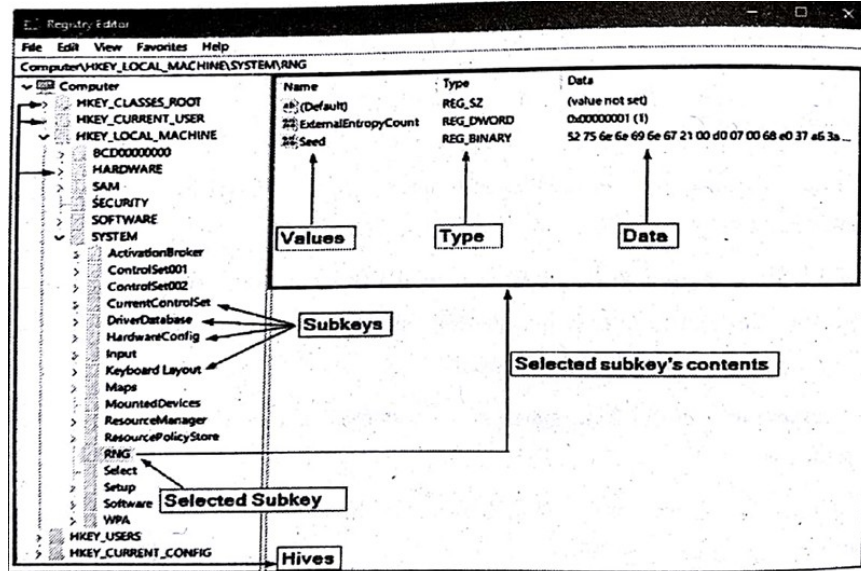
Module 4: Windows and Unix Forensics Investigation

1. [Mid 2] Windows event logs store information about: [1 Mark]
(a) **System activities**
(b) Photos
(c) Music
(d) Videos
2. [Mid 2] The Windows Registry stores: [1 Mark]
(a) **System configuration settings**
(b) Images
(c) Documents
(d) Programs only
3. [Mid 2] Which folder stores temporary files in windows? [1 Mark]
(a) System32
(b) **Temp**
(c) Program files
(d) Documents
4. [Mid 2] Explain the concept of “Trust Relationships”. [3 Marks]

A **trust relationship** is a security connection between systems or domains where one system trusts another for user authentication and access to resources. It allows users already authenticated in one domain to access resources in another without needing to log in again. It is commonly used in Windows environments for resource sharing. However, if one trusted system is compromised, it may lead to unauthorized access to other connected systems.

5. [Mid 2] List and explain different Windows system Artifacts. [3 Marks]
 - **Windows Event Logs:** Store records of system activities such as logins, errors, and security events, useful for tracking user actions.
 - **Windows Registry:** Contains system configuration settings, installed programs, and user activity information.
 - **Temporary Files (Temp Folder):** Stores temporary data created by applications, which may contain traces of user activity.
 - **Prefetch Files:** Keep information about recently executed programs to improve performance and help identify application usage.
 - **Recycle Bin:** Stores deleted files, which can often be recovered and analyzed as evidence.

6. [Mid 2] Draw and explain windows registry structure. [4 Marks]



The **Windows Registry** is a **hierarchical database** used by the Windows operating system to store system configuration, hardware settings, user preferences, and application information.

Components:

1. **Hives:** Top-level components that store major categories of system configuration.
 - **Examples:** HKEY_CLASSES_ROOT, HKEY_CURRENT_USER, HKEY_LOCAL_MACHINE.
2. **Keys and Subkeys:** Keys act like folders, and subkeys are nested inside them, forming a hierarchical structure for organizing data.
 - **Example:**

```
HKEY_LOCAL_MACHINE
├── SYSTEM
│   ├── ControlSet001
│   │   └── HardwareConfig
```
3. **Selected Subkey:** When a subkey is selected in the left panel, its contents appear on the right side of the Registry Editor.
4. **Values:** Entries within a key that store actual configuration settings.
 - **Examples:** (Default), EntropyCount, Seed.
5. **Type:** Indicates the format of the stored data.
 - **Examples:** REG_SZ (String value), REG_DWORD (32-bit numeric value), REG_BINARY (Binary data).
6. **Data:** The actual content of the value, representing system settings.
 - **Example:** EntropyCount = 0x00000001 (1)

7. What is Windows Forensic Analysis?

Windows forensic analysis is the process of collecting, preserving, examining, and analyzing digital evidence from a Windows operating system during a forensic investigation. It involves analyzing system components such as the file system, Windows registry, event logs, user activities, and application data to detect unauthorized access, recover deleted files, and understand system behavior. The main goal is to reconstruct events and gather reliable digital evidence related to cyber incidents or criminal activities.

8. What is a file system?

A **file system** is a method used by an operating system to organize, store, manage, and retrieve data on storage devices such as hard drives, SSDs, and USB drives. It defines how data is structured into files and directories and how these files are stored and accessed on the disk. The file system also keeps track of information such as file names, locations, sizes, permissions, and timestamps. **Common examples** of file systems include FAT, NTFS, exFAT, and EXT.

9. Write a brief note on Reviewing Pertinent Logs in UNIX.

In UNIX systems, **log files provide important information during incident response and forensic investigations**. These logs record various system activities such as user logins, system startups, shutdowns, and network service events. During an investigation, analysts review these logs to identify suspicious activities, system errors, or unauthorized access attempts. **Most UNIX log files are stored in the /var/log directory**, which contains records of system operations and security events. In some UNIX variants, logs may also be found in directories such as /usr/adm or /var/adm. Occasionally, certain log files may be located in other directories like /etc.

10. Write a note on Windows file system?

The **Windows file system** is the structure used by the Windows operating system to store, organize, and manage data on storage devices such as hard drives and SSDs. It controls how files and directories are created, accessed, stored, and retrieved. **Windows mainly uses the NTFS (New Technology File System)**, although older systems also support FAT and FAT32. NTFS provides advanced features such as file permissions, encryption, compression, and journaling, which improve security and reliability. The Windows file system **organizes data into files and folders (directories)**. It also maintains metadata such as file name, size, location, creation time, modification time, and access permissions.

11. Write a note on UNIX file system?

The **UNIX file system** is a hierarchical structure used by UNIX operating systems to organize and manage files and directories on storage devices. It **stores data in a tree-like structure that begins with the root directory (/)**, from which all other directories and files branch out. In the UNIX file system, **everything is treated as a file**, including devices and processes. Important directories include /bin (system binaries), /etc (configuration files), /home (user directories), /var (log files), and /tmp (temporary files). The UNIX file system **manages data using inodes**, which store metadata such as file permissions, owner information, size, and timestamps. This structure helps the operating system efficiently locate and manage files.

12. Write a brief note on Windows Recycle Bin Forensics.

The **Windows Recycle Bin** stores files that have been deleted by users but are not permanently removed from the system. Instead of deleting a file immediately, Windows moves it to the Recycle Bin, allowing it to be restored later. Deleted files are stored in the **\$Recycle.Bin folder** located in the root directory of the drive (e.g., C:\$Recycle.Bin). Inside this folder are subfolders named with the user's Security Identifier (SID), which represent individual user accounts. Each user's folder contains the files they have deleted.

For each deleted file, two types of files are stored:

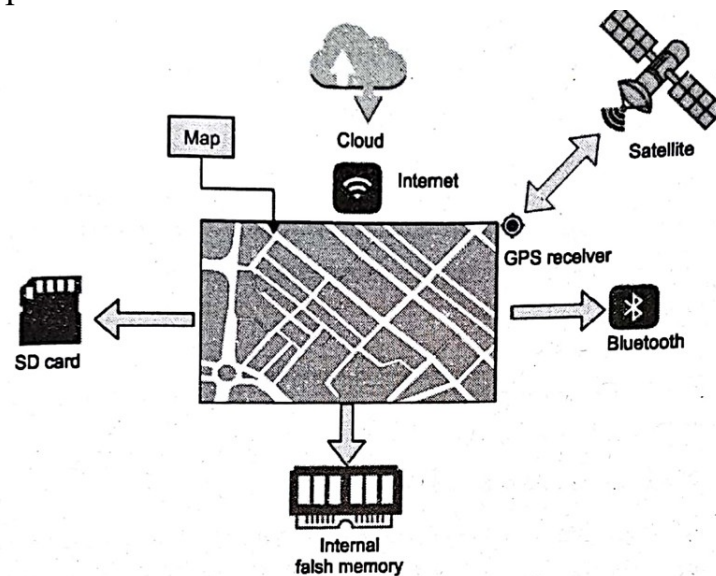
- **\$R files:** Contain the actual deleted data that can often be recovered.
- **\$I files:** Contain metadata about the deleted file, such as the original file path, file size, and deletion time.

During forensic analysis, investigators examine the \$Recycle.Bin directory and the SID folders to identify which user deleted the file. Tools such as \$I Parser can be used to extract metadata from \$I files and generate reports containing information about deleted files.

Module 5: Mobile Forensics

1. [Mid 2] SIM cards can store: [1 Mark]
 - (a) Contacts
 - (b) Messages
 - (c) Network information
 - (d) All of the above
2. [Mid 2] Explain GPS Forensic with diagram. [5 Marks]

GPS forensics is a branch of digital forensics that involves the collection, preservation, and analysis of location data from GPS-enabled devices such as smartphones, navigation systems, vehicles, and wearable devices. It helps investigators determine where a device was located at a specific time and reconstruct movements related to an incident.

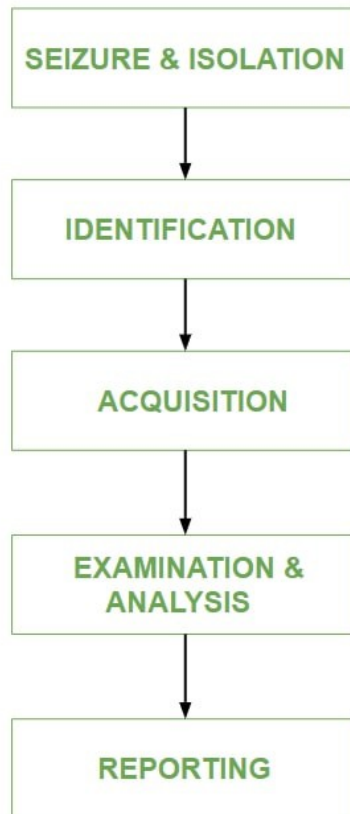


Components:

1. **Satellite:** GPS satellites orbit the Earth and transmit location signals. These signals contain time and positioning information.
2. **GPS Receiver:** Receives signals from multiple satellites. Using these signals, the device calculates its exact geographic location (latitude and longitude).
3. **Map Application:** The calculated location is displayed on map applications used for navigation and tracking.
4. **Internet / Cloud:** GPS data can be transmitted through the internet and stored in cloud services.
5. **Internal Flash Memory:** Location information is stored in the device's internal memory.
6. **SD Card:** Some devices store maps, logs, or navigation data on an SD card.
7. **Bluetooth:** GPS information may also be shared with other devices using Bluetooth.

3. [Mid 2] Explain Mobile Forensic with diagram. **OR** What is Mobile forensic? Why it is needed. [5 Marks]

Mobile forensics is a branch of digital forensics that involves the collection, preservation, analysis, and presentation of digital evidence from mobile devices such as smartphones, tablets, SIM cards, and memory cards. It helps investigators recover data like call logs, messages, photos, GPS data, and application information.



Mobile Forensic Process:

1. **Seizure & Isolation:** Secure the device and isolate it to prevent remote access or data alteration.
2. **Identification:** Identify the device type, data sources, and potential evidence.
3. **Acquisition:** Create a forensic copy of data without modifying the original.
4. **Examination & Analysis:** Extract and analyze relevant data such as logs, messages, and files.
5. **Reporting:** Document findings and present them in a clear, legal format.

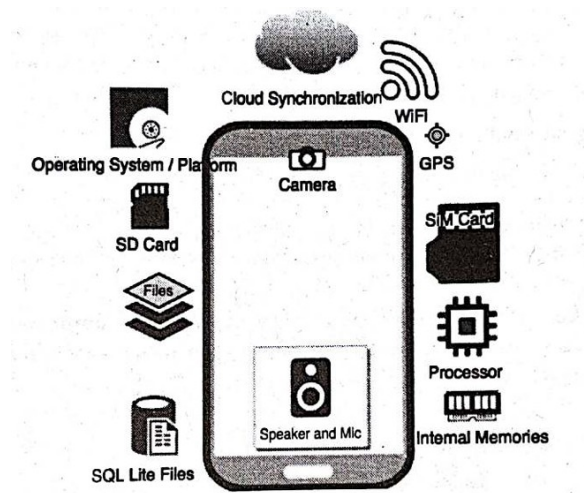
Why Mobile Forensics is Needed:

- Helps in **crime investigation** such as fraud, cybercrime, and terrorism.
- Enables **recovery of deleted data** like messages and files.
- Provides **legal evidence** admissible in court.
- Assists in **tracking user activity** such as calls and location.
- Helps in **communication analysis** between suspects.

4. **[Mid 2]** List and explain at least five data acquisition methods for mobile phones. **OR** Explain any three data acquisition methods for mobile phones. **[3 Marks]**

1. **Manual Acquisition:** Collects data by manually navigating through the phone interface and recording information such as contacts, messages, call logs, and photos.
2. **Logical Acquisition:** Extracts data through the device's operating system using forensic tools. It retrieves files such as contacts, SMS, call logs, and application data but usually cannot recover deleted data.
3. **Physical Acquisition:** Creates a bit-by-bit copy of the entire device memory. It allows investigators to recover deleted files and analyze hidden data.
4. **File System Acquisition:** Extracts the complete file system structure of the device, including directories, files, and system data. It provides more information than logical acquisition but less than physical acquisition.
5. **SIM Card Acquisition:** Extracts data directly from the SIM card, including contacts, SMS messages, and subscriber information stored on the card.

5. **[Mid 2]** Discuss the basic structure of a mobile device. **[4 Marks]**



Components:

1. **Operating System / Platform:** Controls device functions and manages hardware and applications.
2. **Processor (CPU):** Executes instructions and controls overall operations of the device.
3. **Memory and Storage:** Includes internal memory for apps and data, and external storage like SD cards.
4. **SIM Card:** Stores subscriber identity and enables network communication.
5. **Wireless Connectivity:** Includes WiFi, GPS, and cloud synchronization for communication and data access.
6. **Input/Output Components:** Camera, speaker, and microphone used for capturing and interacting with data.
7. **Files and Databases:** Data is stored in files and SQLite databases used by applications.

6. In android terms, what is rooting?

Rooting is the process of gaining administrator (superuser) access to the Android operating system, allowing full control over the device and its system files. It enables users to install custom ROMs, remove pre-installed apps, modify system settings, and run advanced applications. However, rooting can void the warranty, reduce security, and may damage the device if not done properly.

7. What is JTAG?

JTAG (Joint Test Action Group) is a hardware interface used to access and control the internal components of electronic devices such as mobile phones and embedded systems. In digital forensics, it is used as a data acquisition technique to extract data directly from a device's memory, allowing recovery of complete data including deleted or hidden information without relying on the operating system.

8. What are the two most popular applications of GPS?

- **GPS Tracking:** Used to track the location and movement of a device, vehicle, or person.
- **GPS Navigation:** Used to provide directions and route guidance using satellite signals.

9. List four types of evidence that may found within GPS device.

- **Location Data:** Latitude and longitude coordinates showing the device's position.
- **Route History:** Records of routes or paths previously traveled by the device.
- **Waypoints / Saved Locations:** User-saved destinations or frequently visited places.
- **Time and Date Information:** Show when the device was at a specific place.

10. Why SIM card forensics is important?

SIM card forensics is important because it provides valuable information about a mobile user's identity and communication. A SIM card stores data such as IMSI, contacts, SMS messages, call records, and network details. Analyzing this data helps investigators identify users, recover communication history, trace connections between suspects, and collect evidence for criminal investigations. Even if the mobile device is damaged or unavailable, the SIM card can still provide useful digital evidence.

11. What are PIN and PUK numbers?

- **PIN (Personal Identification Number):** A security code used to protect access to a SIM card. It must be entered to unlock the SIM and use services like calls and messages.
- **PUK (Personal Unblocking Key):** A code used to unblock the SIM card if the PIN is entered incorrectly multiple times. It is provided by the mobile network operator to restore access.